

Product Requirements Document: NeuralSync Comprehensive Evaluator (v4.0)

Attribute	Detail
Project Name	NeuralSync Comprehensive Evaluator
Version	4.0 (Comprehensive Strategy)
Owner	[TBD]
Date	November 2025
Core Platform	Single Page Application (SPA) - Fully Responsive
Priority 1 (Mandatory)	Maximum Accuracy: Utilize state-of-the-art psychometric models (CHC, HEXACO, IAT) and Item Response Theory (IRT) scaling.
Priority 2 (Mandatory)	Maximum Ad Engagement: Implement a high-value, high-commitment ad funnel, including a mandatory Gated Result Screen.
Priority 3 (Mandatory)	Beautiful UI: Deliver a modern, dark-themed, and visually rewarding user experience.

1. Executive Summary: Forged in Science, Baked in UI

The NeuralSync Evaluator is a high-value psychometric testing application designed for professional and personal development users. Our competitive edge lies in the scientific rigor of our evaluation (CHC/HEXACO models) combined with highly precise data capture (response latency via Canvas). Monetization is maximized through strategic ad placement that capitalizes on user commitment and engagement, primarily via rewarded video.

Success Metrics

- **Result Accuracy:** User-reported satisfaction score of 9/10 minimum on result relevance.
- **Ad Revenue:** Target 85% Rewarded Video Completion Rate (Slot D).
- **User Experience: Zero Cumulative Layout Shift (CLS)** across all devices.
- **Viral Loop:** Share rate of 12%+ on final results.

2. Scientific Foundation: State-of-the-Art Evaluation Methods

Accuracy is achieved by moving beyond simple Likert scales and implementing tasks that measure implicit bias and cognitive performance under pressure. **Item Response Theory (IRT)** will be used for

final score scaling to provide highly accurate, differentiated results.

A. Cognitive Processing (IQ)

- **Model: Cattell–Horn–Carroll (CHC) Theory** (Focus on Fluid Reasoning (Gf) and Visual Processing (Gv)).
- **Measurement: Adaptive Testing:** Difficulty scales dynamically based on user latency and accuracy.
- **Mandatory Task:** Implement advanced non-verbal **Matrix Reasoning** tasks using the Canvas element for pixel-perfect timing.

B. Emotional Intelligence & Regulation (EQ)

- **Model:** Focus on Affective Empathy and Response Regulation.
- **Measurement: Implicit Association Testing (IAT) principles:** User must rapidly categorize abstract emotional stimuli (words/images) to measure subconscious association speed and bias toward emotional valence.
- **Mandatory Task: Emotional Stroop Test** (color-word interference adapted for emotional words) to track regulation failure under pressure.

C. Core Personality & Behavioral Drivers

- **Model: HEXACO** (Honesty-Humility is essential) and assessment of high-stakes behavioral drivers (Risk/Reward).
- **Measurement: Behavioral Economics Tasks:** Users play a simplified version of the **Balloon Analogue Risk Task (BART)** where precise risk/utility calculations are required, and the Canvas tracks the exact millisecond of their "cash out" decision.
- **Data Integrity:** Raw response latency (ms) for every choice must be recorded and stored for IRT scaling and bias detection.

3. High-Engagement Ad Placement Strategy

Monetization is driven by aggressively capturing the user's high-commitment state at critical transition points. **CLS mitigation is mandatory.**

A. Ad Slot Specification

Slot Name	Location/Trigger	Type	Technical Requirement (CLS Mitigation)	Engagement Strategy
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Slot A: Header	Sticky across all test stages.	Banner (H: 50px)	Must be housed in a fixed, pre-allocated container at all times.	Passive CPM; dynamic content updates based on the current test block (e.g., career ads during cognitive testing).
Slot B: High-Intent Interstitial	Between Block 2 (EQ) and Block 3 (Behavior).	Full-Screen Video/Ad	Dedicated, modal component that prevents any underlying content shift.	Mandatory 3-second delay timer before the "Skip Ad" button appears.
Slot C: Post-Commitment Anchor	Fixed footer, appearing below the Canvas during the <i>last</i> question of each block (Stages 5, 10, 15, 20).	Large Banner (H: 120px)	Fixed minimum height container (h-[120px]). Must not overlap navigation.	Captures attention when the user is mentally "exhaling" after a tough section, before clicking to the next stage.
Slot D: The Gated Result Screen	CRITICAL: Full-screen modal after data processing, before final results.	Rewarded Video/CPA	Full-screen overlay; prevents results display until action is taken.	Highest Revenue. User must choose: 1) Watch 15-second rewarded video (Preferred) OR 2) Wait 30 seconds for non-skippable ad load.

4. UI/UX and Aesthetic Requirements (The "Beautiful UI")

The application must feel sophisticated, secure, and intuitive.

A. Aesthetics and Branding

- **Theme: Dark Mode** is the default and only theme. Use deep blues (#1a202c) and vibrant accent colors (e.g., electric teal or neon green) for scores and interactive elements.
- **Typography:** Use the **Inter** font family. Ensure high contrast (White text on dark backgrounds).
- **Design Language:** Minimalist, highly functional, and visually rewarding. **Heavy use of rounded-x1 and subtle gradients/shadows** to lift interactive elements (buttons, scores) off the dark background.
- **Motion Design:** Use smooth, minimal CSS transitions (transition-all duration-300) for score updates and button hovers.

B. Functional UI Requirements

- **Progress Tracking:** A prominent, persistent progress bar indicating "X of 20 Stages Complete."
- **In-Test Focus:** During testing stages (Blocks 1-4), the UI must be minimal, centering the **Canvas 2D element** to minimize cognitive distraction.
- **Visual Feedback:** Provide immediate, subtle visual feedback (e.g., a brief green pulse for correct, a red glow for incorrect) on every interaction within the Canvas.

- **Result Presentation:** The final result screen must look like a high-value, shareable **"Certificate"**, using a clean structure and advanced data visualizations (simple bar charts or radar plots for domain scores).

5. Detailed Functional Requirements

Require ment ID	Description	Scientific Mandate
FR-S1	The application shall dynamically adjust the difficulty of cognitive tasks (Gf, Gv) based on the running average of the user's last 5 response times.	CHC Model, Adaptive Testing
FR-S2	The app must capture and persist the millisecond latency between stimulus presentation and user response for <i>all</i> 20 stages.	IRT Scaling, Bias Detection
FR-S3	The BART simulation (Block 3) must visually display the calculated probability of failure (risk value) and track the user's risk tolerance setting.	Behavioral Economics, HEXACO
FR-M1	The Gated Result Screen (Slot D) must be impossible to bypass without completing the required ad interaction (Video Watch or 30-sec Wait).	Maximum Ad Engagement
FR-M2	All ad containers (Slots A, B, C) must utilize fixed, reserved screen space to ensure zero CLS .	UX/CLS Mitigation
FR-U1	The final result screen must generate a concise, shareable image/text summary of the user's top three percentile scores.	Viral Loop

6. Technical Requirements & Architecture

The application will be built as a self-contained Single Page Application (SPA) using React for component structure, Firebase Firestore for data persistence, and Canvas 2D for high-precision game mechanics.

A. Tech Stack

- **Frontend:** React (JSX/TSX), Tailwind CSS (for rapid styling and responsiveness).
- **Game Engine:** HTML `<canvas>` + Vanilla JavaScript/React Hooks for high-precision timing and input tracking.

B. Data Model (Firestore)

- **Private User Sessions:** All raw session data (inputs, latency, scores) is stored privately.
 - **Collection Path:** `/artifacts/{__app_id}/users/{userId}/sessions`
 - **Document Structure:** Contains `sessionID` , `userID` , `timestamp` , `rawResponses` :
`[[{stage, choice, latency_ms}], finalScores: {iq, eq, hexaco, percentile} .`

- **Initialization & Authentication:** Firebase services are initialized on component mount, and the user is authenticated using `__initial_auth_token` or `signInAnonymously()` as a **fallback**.

C. Responsiveness

- The entire application, especially the Canvas element, must be **fully responsive**, resizing gracefully to fit mobile, tablet, and desktop viewports without horizontal scrolling. Use Tailwind's responsive prefixes (`sm:` , `md:` , `lg:`).

7. Non-Functional Requirements

- **Performance:** All visual elements must load under 1 second. Canvas tasks must maintain a minimum of 60 FPS on modern devices.